MILITARY SPECIFICATION

# HEATERS, SPACE: MULTIFUEL, WITH BLOWER

## 60,000 BTU/HR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers liquid-multifuel-burning space heaters having 60,000 Btu/hr capacity.

1.2 Classification. The heaters shall be of the following types, as specified (see 6.2).

Type I - 120-volt ac, 50/60 Hertz, single phase.
Type II - 208-volt ac, 400 Hertz, three phase.

1.2.1 Part number. The military specification part numbers for the heaters shall be as follows:

Part No. M11511-1 for type I heaters, 120 volt, 50/60 Hertz, single phase.
Part No. M11511-2 for type II heaters, 208 volt, 400 Hertz, three phase.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 4520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-17T18:17Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| CC-M-1807 | - Motor, Alternating-Current Fractional and Integral Horsepower. |
| GG-P-455 | - Plate, Photographic (Photosensitive Aluminum). |
| QQ-S-781 | - Strapping, Steel, and Seals. |
| VV-F-800 | - Fuel Oil, Diesel. |
| VV-G-1690 | - Gasoline, Automotive, Leaded or Unleaded. |
| PPP-B-566 | - Boxes, Folding, Paperboard. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-F-320 | - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes. |
| PPP-P-40 | - Packaging and Packing of Hand Tools. |
| PPP-T-45 | - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing. |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-V-173 | - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment). |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-G-3056 | - Gasoline, Automotive, Combat. |
| MIL-W-5086 | - Wire, Electric, Hookup and Interconnecting Polyvinyl Chloride-Insulated, Copper or Copper Alloy Conductor. |
| MIL-B-5423 | - Boot, Dust and Water Seal (For Toggle and Push-Button Switches and Rotary-Actuated Parts), General Specification for. |
| MIL-F-5591 | - Fasteners; Panel. |
| MIL-C-5809 | - Circuit Breakers, Trip-Free, Aircraft, General Specification for. |
| MIL-R-6106 | - Relays, Electric, General Specification for. |
| MIL-B-7883 | - Brazing of Steels, Copper, Copper Alloys, Nickle Alloys, Aluminum and Aluminum Alloys. |
| MIL-R-11050 | - Rectifiers, Metallic, Selenium. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| MIL-P-15024 | - Plate, Tags and Bands for Identification of Equipment. |
| MIL-W-16878 | - Wire, Electrical, Insulated, High Temperature. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-101 | - Preservation, Packaging, and Packing Materials; Test Procedures. |
| FED-STD-595 | - Colors. |
| FED-STD-H28 | - Screw-Thread Standards for Federal Services. |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking for US Military Property. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment. |
| MIL-STD-781 | - Reliability Tests: Exponential Distribution. |
| MIL-STD-810 | - Environmental Test Methods. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |
| MIL-STD-1472 | - Human Engineering Design Criteria for Military Systems, Equipment and Facilities. |
| MIL-STD-1474 | - Noise Limits for Army Materiel. |
| MS3100 | - Connector, Receptacle, Electric, Wall Mounting, Solder Contacts, AN Type. |
| MS3106 | - Connector, Plug, Electric, Straight, Solder Contacts, AN Type. |
| MS51085 | - Filter, Fluid, Pressure, Automotive Fuel (10 GPH Filtration). |
| MS51086 | - Strainer, Sediment, Automotive Fuel (10 GPH). |
| MS51321 | - Pump, Fuel, Electrical - 24 Volts DC 25 G.P.H. Capacity. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Power Test Code.
Supplement on Instruments and Apparatus, Part 10, Flue Exhaust Gas Analysis.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AIR MOVEMENT AND CONTROL ASSOCIATION, INC.

AMCA Standard 210 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the Air Movement and Control Association, Inc. 30 West University Drive, Arlington Height, IL 60004.)

SOUTHERN CALIFORNIA - Air Pollution Control District

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS

ASHRAE Handbook and Product Directory - 1977 Fundamentals

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Description. The heaters shall be as shown in figures 1 through 3 and as specified herein. The heaters shall be of the liquid-multifuel-burning, uncontaminated, forced hot-air type. The heater shall have a measured output heating capacity of not less than 60,000 British thermal units per hour (Btu/hr) at sea level (0 to 500 feet) and an ambient temperature of 70° F plus or minus 10° F and with the airflow specified in 3.12. The heater shall perform as specified herein at an altitude of 5,000 feet. The heater shall include a fuel system, an ignition system, a combustion system, a circulation-air system, an electrical system, all necessary controls and wiring and a sheet-metal enclosure. The weight of the heater with all its components installed (not including the remote fuel pump) shall be not more than 150 pounds. The heater shall operate as specified herein when rotated in 90-degree increments about an axis parallel to the long dimension of its base. Interchangeable mounting brackets shall be provided for mounting the heater in any of the 90-degree increments. Unless otherwise specified (see 6.2), heaters shall be delivered for mounting in the air-discharge down position. No component, except the exhaust nipple, control box, mounting bracket, fuel connections, fasteners, and carrying handles, shall extend beyond the dimensional limits of the heater case. The types I and II heaters shall operate from the following power source, respectively: 120 volt ac, 50/60 Hertz, single phase, and 208 volt ac, 400 Hertz, three phase. On type I heaters, the capacity specified herein shall be when operating on 60-Hertz power; a 15 percent loss shall be allowed when operating on 50-Hertz power.

3.2 First-article (preproduction model). The contractor shall furnish four heaters for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce heaters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more heaters (see 6.2) for inspection as specified in 4.4.

3.4 Maintainability. The heater shall permit accessibility to all parts for servicing, maintenance, and repair. Replacement and adjustment of parts shall



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

be accommplished with minimum drainage of fuel and with minimum interference from, injury to, or removal of other parts or wiring. Ease-of-maintenance provisions shall incorporate, to the maximum extent practicable, features insuring operating clearances and facilitating applicable maintenance and servicing operations at low temperatures. Covers or plates that must be removed for adjustment or repair shall be equipped with captive, quick-disconnect fasteners. Service adjustments required shall be the minimum consistent with required performance. Accessible adjustments shall be quickly returnable to their original settings. Design of the heater shall eliminate where possible the assembly of wrong combinations of parts and shall eliminate or minimize the holding of heavy or awkward parts while being brought into final position and while being fastened. Minimum allowable time between scheduled organizational maintenance services shall be 125 hours. The mean-time-to-repair shall be 2 hours at the organizational level, 8 hours, direct support and 24 hours at the general support level. The maintenance ratio of total scheduled and unscheduled maintenance manhours to total operating hours shall not exceed 0.05.

3.5 Material. Material shall be as specified herein. To the maximum extent possible recycled and recovered materials should be used in lieu of virgin raw materials. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free from defects or imperfections which might affect the performance, maintainability, or appearance of individual components or, the finished heater assembly. Materials subject to atmospheric corrosion or rust shall not be treated with a temporary treatment such as grease or oil in order to meet environmental condition requirements. None of the above shall be interperted to mean that the use of used or rebuilt products will be allowed.

3.6 Environmental conditions.

3.6.1 Operating temperatures. In addition to starting and operating under the conditions specified in 3.1, the heater shall start and operate without failure or malfunction under sea level conditions at ambient temperatures between minus 50° and plus 80° F.

3.6.2 Storage temperature. The heater shall not be damaged nor its operation impaired by storage at an ambient temperature of plus 155° F for periods up to 4 hours daily, no solar radiation; and at minus 65° F for periods up to 24 hours, no solar radiation.

3.6.3 Tilted operation. The heater shall start and operate without failure or malfunction when positioned at any angle from horizontal to 10 degrees from horizontal in any plane.

3.6.4 Shock and vibration. The heater shall withstand and absorb shock and vibration as specified herein without structural damage, visual misalignment of components, malfunction of components, leaks, abnormal vibration, or any other irregular operation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

3.6.4.1 Shock. The heater shall withstand three shocks applied along each of the three mutually perpendicular directions that are respectively parallel to the edge of the heater (a total of 15 shocks). Each shock shall be not less than 15 g's (g = acceleration caused by gravity) and shall have a duration of not less than 11 milliseconds. The shock wave form shall be as required by figure 516.2-2 of MIL-STD-810.

3.6.4.2 Vibration. The heater shall withstand vibration as specified in MIL-STD-810, method 514.2, category F, procedure VIII, table 514.2-VI, curve V. The test time shall be 60 minutes total per plane.

3.6.5 Salt fog and humidity. When subjected to the tests specified herein, the heater shall show no evidence of peeling of paint, malfunctioning, rough operation, or electrical shorts and shall show no evidence of deterioration, corrosion, or change in tolerance limits of any internal or external parts which could prevent the unit from meeting operational and maintainability requirements.

3.6.5.1 Salt fog. The heater shall withstand salt-fog exposure as specified in MIL-STD-810, method 509.1, procedure I.

3.6.5.2 Humidity. The heater shall withstand humidity exposure as specified in MIL-STD-810, method 507.1, procedure I.

3.6.6 Sound level. When subjected to the test specified in 4.6.2.5, the sound level of the heaters shall conform to MIL-STD-1474, category E for steady-state noise.

3.6.7 Endurance. The heater shall be capable of operating for 6,000 cycles, with each cycle consisting of 20 minutes at full heat (on-time) and 10 minutes at no heat (off-time). The 3000 hours accumulated from these cycles shall include 2000 on-time hours and 1000 off-time hours. During this endurance operation (see 4.6.2.8), the heat output shall not vary more than 10 percent from that obtained in the capacity test (see 4.6.2.1). Except for refueling, the heater shall operate without maintenance for not less than 250 hours. At the end of 250 hours, cleaning of the burner head, cleaning of the fuel filter(s) and cleaning of the ignition plug(s) shall be permitted. The ignition plug(s) and fuel filter(s) may be replaced every 1000 hours.

3.7 Safety. All exposed parts, except the heater exhaust stack, that are subject to high operating temperatures or are energized electrically shall be insulated, enclosed, guarded, or grounded. All moving parts that are of such nature or so located as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions.

3.8 Efficiency. The heater shall have an indicated efficiency of not less than 70 percent and shall have a measured efficiency of not less than 65 percent. The indicated efficiency shall be based on the flue-gas analysis, and the measured efficiency shall be based on the measured out-put.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

3.9 Fuel system. The heater shall operate with gasoline conforming to MIL-G-3056 or VV-G-1690 and any other gasoline up to 100 octane; and with diesel fuel conforming to VV-F-800, class DF-1, DF-2 and DF-A having a cloud point not lower than minus 55° F. The fuel system shall be designed for operation when connected to one external fuel line. Any excess fuel pump capacity shall be handled by a pressure regulating or bypass arrangement inside the heater case. A fuel overflow line shall be used with a float control for venting and for overflowing fuel in the event of a malfunction. The fuel overflow line shall not be used as a return fuel line for the excess capacity of the fuel pump or any continual overflow from the fuel system. The fuel system shall include a fuel line connection mounted through the heater case and shall terminate with a 1/4 inch brass, flare-type fitting for connecting the main fuel line from the fuel supply heater case with the same type fitting as specified for the fuel line connection. A flare-type cap, with captive chain, shall be provided with each external fitting. A fuel filter conforming to MS51085 or MS51086 shall be provided within the heater case to filter all fuel. The fuel-supply tank and external fuel lines shall not be furnished. The fuel system shall not leak and shall completely shut off fuel flow when the heater is not in operation so that fuel pressure cannot force fuel into the combustion chamber.

3.9.1 Remote fuel pump. When specified (see 6.2), a remote fuel pump conforming to MS51321-1 shall be furnished for installation in a main fuel line. A remote fuel pump electrical connector conforming to MS3100 shall be recess-mounted on all heaters (see left end view figure 1) to provide 24-volt dc electrical power for the pump. The 24 volt dc electrical power shall be supplied by the heater, and the pump shall be controlled by the thermostat. A remote fuel pump shall be supplied with each preproduction model heater. An electrical connector mating with the MS3100 connector mounted on the heater case shall be supplied with all heaters. Additional fuel pressure, if required to operate the fuel burner (such as a high pressure gun type burner), shall be supplied by a separate fuel pump located inside the heater case.

3.10 Ignition system. The ignition system shall ignite the fuel air mixture and maintain combustion under all conditions specified herein. The ignition system shall consist of an electrode or electrodes located at the fuel nozzle. Power shall be furnished for the electrode(s) through transformers. The ignition wire(s) between the electrode(s) and the transformer shall be equipped with threaded connectors on both ends of the wire.

3.11 Combustion system.

3.11.1 Combustion-air supply. The combustion-air supply blower shall supply air at a pressure to the burner head for efficient operation of the burner at outputs and all operating conditions specified herein. The blower shall be driven by a dripproof, fungusproof, permanently lubricated motor conforming to CC-M-1807. The motor shall be equipped with a thermal-overload protector to prevent motor operation above the safe operating temperature for which the motor is rated. Brush-type motors, when applicable shall be provided with readily accessible access opening for the inspection and renewal of brushes and springs,



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

or shall be capable of lasting the full 2,000 hours of burning time of the service life test. Blower operation shall be continued after each heating cycle to purge the heater.

3.11.2 Burner. The burner shall provide the fuel-air ratio required for maintaining continuous combustion with not more than 0.02 percent carbon monoxide in the flue gases. The burner shall not become clogged by the products of combustion. The burner shall be located and attached to the combustion chamber so as to facilitate removal of the burner components for inspection, replacement, and adjustment. Burner operation shall not be hindered by the back pressure induced by a flue-gas exhaust pipe equivalent to the resistance of four, 2-inch, 90-degree elbows and 6 feet of 2-inch diameter pipe.

3.11.3 Combustion chamber and heat exchanger. The combustion chamber, heat exchanger and exhaust fitting shall be constructed of steel. The burner and combustion chamber shall be attached to, or be part of, the primary pass of the heat exchanger. When used to extract heat, baffles, or fins shall be continuously bonded to the heat exchanger by brazing or some other method that will withstand the temperatures encountered. The combustion chamber and heat exchanger shall be sealed to prevent the products of combustion from contaminating the circulated air. The exhaust end of the heat exchanger shall heater housing not less than 1-1/4 inches. The heat exchanger shall not leak when subjected to 5 pounds per square inch gage (psig) air pressure with all external openings sealed.

3.12 Circulation-air system. The circulation-air system shall consist of a motor-driven blower, dampers, inlet and discharge openings, and when specified (see 6.2), a deflector hood. When the heater is operating in the heat mode at conditions of sea level and 70° F (see 3.1), the circulating-air discharge shall be not less than 38 pounds per minute (lbs/min) against an external static pressure of 0.20 inch of water, and not less than 32 lbs/min against an external static pressure of 0.35 inch of water at sea level. On type I heaters the above specified capacity shall apply when operating on 60-Hertz power.

3.12.1 Circulation-air blower. The circulation-air blower shall be motor-driven. The blower shall operate without damage to any of the components when the external static pressure has been removed. The blower shall be fitted with a 2-inch IPS threaded nipple which shall extend through the driven by a dripproof, fungusproof, permanently lubricated motor, conforming to CC-M-1807. The motor shall be equipped with a thermal over-load protector to prevent motor operations above the safe operating temperature for which the motor is rated. Brush-type motors, when applicable, shall be provided with readily accessible access openings for the inspection or renewal of brush or spring. The blower circuit shall include controls arranged so that when the heater is started, operation of the circulating air blower shall be delayed until the temperature in the heat exchanger assures continued combustion of the fuel-air mixtures. When the minimum temperature for continued operation is reached, the blower shall begin operation by an automatic control. Similarly, an automatic control shall continue the blower operation following cessation of combustion until residual heat has been purged from the heater.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

3.12.2 Dampers. Dampers shall be provided at the two air inlet openings on each side for regulating the amount of recirculated and fresh makeup air passing over the heat exchanger. The dampers shall be interconnected and shall be manually operated by a common damper control so that opening of one damper will automatically close the other damper by the same ratio.

3.12.3 Deflector hood. When applicable (see 3.12), deflector hoods shall be as shown on figure 3, and as specified herein. The hood shall be constructed of sheet metal having a thickness of not less than 0.024 inch. The hood shall be a component separate from the heater case, but fastened to the air-discharge opening so as to be removable.

3.13 Electrical system and controls. All controls, switches, and junction blocks shall be compatible with the operating voltage specified herein. All internal wire shall conform to MIL-W-5086 or MIL-W-16878, type B, except where required for high-voltage spark wires, transformer leads, and radio-interference reduction. Any selenium rectifiers employed shall conform to MIL-R-11050. Electrical relays shall be hermetically sealed and shall conform to MIL-R-6106. The electrical system and heater case shall be protected by a ground connected to the ground terminal of the power-supply connection. The control system shall include a heat output control, on-off switch, safety controls, and control panel.

3.13.1 Heat output control. A room thermostat shall be provided for remote mounting in the space being heated. The thermostat shall control the burner operation within plus or minus 1-1/2° F of the dial setting. The thermostat shall be of the heavy duty type designed to withstand frequent cycling (15 to 20 cycles per hour) and shall operate in any position. The temperature range shall be from plus 55° F or less to plus 85° F or more, and the selector dial shall be so calibrated. The thermostat case shall be metal, and all non-metal parts shall be protected against breakage. A receptacle with mating plug shall be furnished on the control panel to provide for electrical connection and electrical ground between the room thermostat and the heater.

3.13.2 Safety controls.

3.13.2.1 High-temperature. A nonadjustable, automatic control shall shut off the fuel supply to the combustion chamber in the event of high temperatures detrimental to components within the heater or before the air discharge temperature exceeds 275° F. The control shall be an automatic reset type, and shall protect the heater from excessive temperature when operated in all positions specified in 3.1.

3.13.2.2 Flame-failure. An automatic control shall shut off the fuel supply to the combustion chamber and stop all heater operation in the event of either ignition failure or flame failure. At normal ambient temperatures (70°F) the fuel shutoff shall be accomplished within 60 seconds after failure has occurred. At low ambient temperatures (-50°F) the fuel shut-off shall be accomplished within 120 seconds after the failure has occurred. A manual reset shall be provided.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

3.13.3 Control box. The control box shall be as shown in figure 2, shall be mounted on the heater case and shall be capable of being remotely located from the heater without alteration. The control box shall contain the on-off switch, the indicating lights for operation and failure, operating instructions, and any other controls necessary for the remote operation of the heater. The control box shall be fitted with a polarized electrical fitting to provide electrical connection between the control box and the heater. The locations of controls and receptacles on and within the control box may be varied from the locations shown in figure 2.

3.13.3.1 On-off switch. The on-off switch shall be of the three-position toggle type and shall control the operation of the heater as follows:

(a) With the switch positioned in the "on" position, the combustion cycle and the heater control system shall be energized.
(b) With the switch positioned in the "fan" position, only the circulation-air blower shall operate.
(c) With the switch positioned in the "off" postion, the combustion cycle and the heater control system shall be de-energized. The center position shall be "off".

3.13.4 Receptacles.

3.13.4.1 Power connection. A power receptacle conforming to MS3100, part number MS3100-R18-10P, with cap and chain shall be flush mounted in the heater case. A mating plug conforming to MS3106, part number MS3106-E18-10S, shall be furnished with each heater.

3.13.4.2 Thermostat receptacle. A three-contact receptacle conforming to MS3100, part number MS3100-R14S-7S, with cap and chain, shall be flush mounted in the heater control box. A mating plug conforming to MS3106, part number MS3106-E14S-7P, shall be furnished with each heater.

3.13.4.3 Auxiliary fuel-pump electrical connector. A receptacle conforming to MS3100, part number MS3100-R14S-9S with cap and chain, shall be flush mounted in the front of the heater case. A mating plug conforming to MS3106, part number MS3106-E14S-9P, shall be furnished with each heater.

3.13.5 Electrical protective devices. An automatic circuit breaker shall be provided, conforming to MIL-C-5809, to protect the electrical equipment from damage caused by overload of the electrical system due to short or open circuits or to failure of circuit parts. The circuit breaker shall be accessible so as to allow both removal and resetting from the front of the heater. The circuit breaker shall have a dust and waterproof boot(s) conforming to MIL-B-5423.

3.13.6 Electrical characteristics. The heaters shall start and operate on the electrical power input characteristics as follows:

Type I

(a) 103.5 to 126.5 volts, single phase, 60 Hertz (Hz).
(b) 110 to 120 volts, single phase, 50 Hz to 63 Hz.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

Type II

(a) 188 to 228 volts, 3 phase, 400 Hz.
(b) 198 to 218 volts, 3 phase, 380 to 420 Hz.

3.14 Structural components. Standard structural shapes shall be used. Material for structural shapes shall conform to ASTM A36.

3.15 Heater case. The heater case shall be constructed of sheet metal having a thickness of not less than 0.049 inch. All removable panels and access doors, with the exception of the return air opening cover plates, shall be provided with fasteners conforming to MIL-F-5591, type II, class and style optional. Both side panels shall be removable without the removal of any other component. The heater case shall be provided with access doors, removable panels, and openings as required for operation and maintenance and as specified herein. Cover plates and provision for attachement shall be provided for all return-air openings. The heater shall be shipped with the bottom return air cover plate installed on the heater. The side cover plates shall be detached and overpacked with the heater.

3.16 Carrying handles. Carrying handles shall be provided on each end of the heater as shown in figure 1. Each handle shall withstand 2-1/2 times the resulting vectorial force imposed when the heater is suspended vertically by the handles in the attitude shown in figure 1. The handles shall be capable of being extended out from the heater for carrying the heater and placed against the heater for storage and operation. The handles shall be of such size and shape to afford gripping with artic type mittens.

3.17 Fasteners. All steel fasteners shall be plated with zinc, or nickle or resist atmospheric corrosion under the environmental conditions specified herein. Sheet metal or self-tapping screws shall not be used. Threads shall conform to FED-STD-H28.

3.18 Electromagnetic interference. The electromagnetic interference emission characteristics of the heaters shall conform to MIL-STD-461A, Notice 1, table A-1, Equipment Class-Tactical Non-Communications Electronic Equipment. Specific subtests of MIL-STD-461A shall be RE02 over frequency range from 0.15 to 1000 MHz and ACE-1 over frequency range from 1.5 to 65 MHz.

3.19 Lubrication.

3.19.1 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with grease conforming to MIL-G-23827. The heater shall be assembled, run-in (broken-in), tested, and delivered with this grease. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other grease. A tag shall be attached in a conspicuous place to indicate which military grease has been used.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

3.19.2 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.20 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.20.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.20.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the supplier shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.21 Identification marking. The heater shall be identified in accordance with MIL-STD-130. The marking shall be applied to the heater on identification plates conforming to MIL-P-15024/6, type A or H, style III, size 12. The ownership designation shall be "U.S."

3.21.1 Other markings. The heater shall be equipped with markings identifying items that are involved in its operation and handling such as controls, fuses, and input power connections.

3.22 Instruction plates. Each heater shall be equipped with instruction plates including warnings, diagrams, and cautions conspicuously describing any special or important procedures to be followed in operating and servicing the heater. The instruction plates shall conform to MIL-P-514, compositon A, class 2 or composition C with materials in accordance with GG-P-455, type optional, grade A, class 1, and shall be installed on the control box door. The words "Warning" and "Caution" shall be red, matching color No. 31136 of FED-STD-595.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

3.23 Schematic and wiring diagram. A plate or plates containing a schematic diagram and wiring diagram shall be mounted on the inner surface of the heater panel access door. The diagram shall portray the complete operational and functional sequence of the circuit for analysis and maintenance of the heater. The wiring diagram shall portray the physical location of all wiring interconnections in the same relationship as when they are installed. Identification markings of wires, instruments, control devices, circuit breakers, fuses, and connections shall be shown.

3.23.1 Selenium rectifier caution plates. When selenium rectifiers are used (see 3.13), a caution plate shall be conspicuously located adjacent to each rectifier or group of rectifiers and shall state the following: "If malfunctioning of the rectifier occurs, do not handle the damaged rectifier while it is hot. To avoid inhalation of the poisonous fumes or absorption of poisonous selenium compound through the skin, the rectifier compartment should be ventilated thoroughly before the rectifier chassis is removed."

3.24 Treatment and painting. The portions of the heater normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.25 Human factors. The characteristics of the heater shall provide for operation by personnel in all type clothing and shall be designed in accordance with section IV of MIL-STD-1472.

3.26 Reliability. The specified Mean-Time-Between-Failures (MTBF) shall be 480 hours when tested as specified in 4.6.2.9. The lower MTBF shall be 320 hours.

3.27 Workmanship.

3.27.1 Metal fabrication. Metal used in fabrication shall be free from kinks ınd sharp bends. The straightening of material shall be done by methods that ;ill not cause injury to the material. Corners shall be square and true. Flame ;utting, using tips suitable for the thickness of the steel, may be employed nstead of shearing and sawing. All bends shall be made with controlled means o insure uniformity of size and shape. Precaution shall be taken to avoid verheating. Heated steel shall be allowed to cool slowly. External surfaces hall be free of burrs, sharp edges and corners, except when sharp edges or orners are required or where they are not detrimental to safety.

3.27.2 Brazing. Brazing of steels, copper alloys and nickel alloys shall onform to MIL-B-7883.

3.27.3 Bolted connections. Boltholes shall be accurately formed and shall ve the burrs removed. Washers or lockwashers shall be provided where cessary. Matching thread areas securing bolts conforming to SAE J429 or pscrews shall be sufficient to withstand the tensile strength of the bolt. l fasteners shall be correctly torqued and shall have full thread engagement.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

3.27.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.27.5 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Qualification, Section 5.
Welding Qualifcations of the ASME, Section IX.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, contractors standard welder qualification may be subsituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.27.6 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the heater to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the heater shall be at the option of the contractor, and shall conform to the provisions of the American Welding Society Publication D1.1 Structural Steel Welding Code or ASME Boiler and Pressure Code Section VIII.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirments as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction heaters shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. One preproduction heater shall be subjected to the tests marked "X" in column 2a of table I, and the remaining three preproduction heaters shall be subjected to the tests marked "X" in column 2b of table I. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production heaters will be selected at random by the Government from the heaters being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in column 4 of table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production heater shall not exclude the remaining heaters from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of any initial production heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the inital production heaters and shall be cause for refusal by the Government to continue acceptance of production heaters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on heaters previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all heaters accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each heater shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each heater shall be subjected to the test marked "X" in column 3 of table I. Failure of any test shall be cause for rejection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

TABLE I. Test-schedule. 1/ 2/

| Type of Test | | | | | | | |
|---|---|---|---|---|---|---|---|
| Test sequence | Preproduction Test | | Quality Conformance Test | Initial Production Test | Test | Quality Assurance Paragraphs | Requirement Paragraphs |
| 1 | 2a | 2b | 3 | 4 | 5 | 6 | 7 |
| 1 | X | X | X | - | Pressure | 4.6.2.11 | 3.11.3 |
| 2 | X | X | - | X | Controls | 4.6.2.4 | 3.13 |
| 3 | X | X | - | X | Capacity and Airflow | 4.6.2.1 | 3.1, 3.8, 3.11.2, 3.12 |
| 4 | X | - | - | X | Operating position variation | 4.6.2.6 | 3.1 |
| 5 | X | - | - | X | Tilted operation variation | 4.6.2.6.1 | 3.6.3 |
| 6 | X | - | - | - | Voltage and frequency variation | 4.6.2.15 | 3.13.6 |
| 7 | X | - | - | X | Sound level | 4.6.2.5 | 3.6.6 |
| 8 | X | - | - | - | Altitude | 4.6.2.2 | 3.1, 3.8 |
| 9 | X | - | - | X | Electonmagnetic Interference | 4.6.2.10 | 3.18 |
| 10 | X | - | - | - | High Temperature | 4.6.2.3.1 | 3.6.1, 3.6.2 |
| 11 | X | - | - | X | Low Temperature | 4.6.2.3.2 | 3.6.1, 3.6.2 |
| 12 | X | - | - | X | Humidity | 4.6.2.3.3 | 3.6.5 |
| 13 | X | - | - | - | Salt-fog | 4.6.2.3.4 | 3.6.5 |
| 14 | X | - | - | X | Vibration | 4.6.2.3.5 | 3.6.4 |
| 15 | X | - | - | - | Shock | 4.6.2.3.6 | 3.6.4 |
| 16 | X | - | - | - | Carrying Handle | 4.6.2.7 | 3.16 |
| 17 | X | - | - | X | Motor Locked Rotor | 4.6.2.13 | 3.12.1 |
| 18 | X | - | - | X | Combustion Safety | 4.6.2.14 | 3.13.2 |
| 19 | - | X | - | - | Endurance | 4.6.2.8 | 3.6.7 |
| 20 | - | - | X | - | Individual Performance | 4.6.2.12 | 3.1,3.8 3.11.2, 3.13 |
| 21 | - | - | - | X | Human Factors | 4.6.2.17 | 3.25 |
| 22 | - | - | - | X | Reliability | 4.6.2.9 | 3.26 |
| 23 | - | - | - | X | Maintenance Evaluation | 4.6.2.18 | 3.4 |
| 24 | - | - | - | X | Safety | 4.6.2.16 | 3.7 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

1/ Perform test marked "X".
2/ Tests shall be performed in the sequence indicated, or as approved by the Contracting Officer.

4.6 Inspection procedure.

4.6.1 Examination. The heater shall be examined for the characteristics listed in table II.

TABLE II - Examination schedule.

| Number | Characteristic | Requirement Paragraph |
|---|---|---|
| Major | | |
| 101. | Total weight of the heater with all its components installed exceeds 150 pounds. | 3.1 |
| 102. | Dimensions not as specified. | Figure 1 |
| 103. | Components other than the exhaust nipple, control panel, mounting brackets, fuel connections, fasterners, or carrying handles extended beyond the dimensional limits of the heater case. | 3.1 |
| 104. | Openings and cover plates not as specified. | Figure 1 |
| 105. | Dissimilar metals not isolated or not treated as specified. | 3.1 |
| 106. | Fasteners not as specified. | 3.17 |
| 107. | Ease of maintenance provisions not as specified. | 3.4 |
| 108. | Material not as specified. | 3.5 |
| 109. | Parts hazardous to operating or maintenance personnel not insulated, enclosed, guarded, or grounded when required. | 3.7 |
| 110. | Fuel system leaks. | 3.9 |
| 111. | Remote fuel pump not as specified. | 3.9.1 |
| 112. | The combustion air supply blower not as specified. | 3.11.1 |
| 113. | Burner(s) not as specified so as to facilitate removal of the burner components for inspection, replacement, and adjustment. | 3.11.2 |
| 114. | Motors not equipped with thermal overload protectors. | 3.12.1 |
| 115. | Dampers not provided as specified. | 3.12.2 |
| 116. | Deflector hood (when applicable) not furnished as specified. | 3.12.3 |
| 117. | Power receptacle not as specified; mating plug not furnished as specified. | 3.13.4.1 |
| 118. | Thermostat receptacle not as specified; mating plug not furnished as specified. | 3.13.4.2 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

TABLE II - Examination schedule.

| Number | Characteristic | Requirement Paragraph |
|---|---|---|
| Major | | |
| 119. | Standard structural shapes not as specified. | 3.14 |
| 120. | Heater case not constructed as specified. | 3.15 |
| 121. | Carrying handles not provided as specified. | 3.16 |
| 122. | Lubricated components not tagged to indicate the temperature range and grade of lubricant used. | 3.19.1, 3.19.2 |
| 123. | Identification marking not as specified. | 3.21 |
| 124. | Other markings not as specified. | 3.21.1 |
| 125. | Instruction plates not as specified. | 3.22 |
| 126. | Schematic and wiring diagram not as specified. | 3.23 |
| 127. | Selenium rectifier caution plates not as specified. | 3.23.1 |
| 128. | Treatment and painting not as specified. | 3.24 |
| 129. | Workmanship not as specified. | 3.27 |

4.6.2 Tests. Unless otherwise specified herein, tests shall be conducted with the heater operating in the air discharge down position, utilizing all components in place. Type I heaters shall utilize 60 Hertz power unless otherwise specified herein. All operating tests shall be conducted with the heater burning a type of gasoline or diesel fuel specified in 3.9. The heater shall be lubricated with military lubricants prior to testing. All instruments shall be calibrated prior to tests.

4.6.2.1 Capacity and airflow. Selection of heaters and conditions for capacity and airflow subtests shall be accomplished in accordance with table III. After all test apparatus and instrumentation has been connected to the heater, operate the heater for not less than 1 hour at test conditions to insure temperature stabilization and uniform operation. Following stabilization, data upon which results will be based shall be recorded every 15 minutes for a total of not less than five complete sets of data. Except for fuel consumption, results shall be based on the average of the five sets of data. Fuel consumption shall be based on the total weight of fuel consumed within the 1-hour data-recording period. Conduct separate 1-hour recording periods as indicated in table III. Each set of data shall consist of not less than the following:

(a) Air in, ° F (read at not less than four locations spaced across the return-air inlet to the heater).
(b) Air out, ° F (read at not less than four locations spaced across the warm-air discharge from the heater and so as not to receive radiant heat).



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

(c) Fuel weight, pounds.
(d) Flue-gas analysis, percent $CO_2$, $O_2$, and CO by volume (samples taken from heater exhaust stack at a point 18 inches from the heat exchanger).
(e) Flue-gas temperature, °F (read within 1 inch of where flue-gas analysis sample is taken).
(f) Static air back pressure, inches of water (read at warm air discharge opening from heater).

(g) Static air pressure and velocity air pressure, inches of water (as required to determine airflow with airflow measuring apparatus being employed).
(h) Power inlet characteristics, volts watts, and amperes.
(i) Barometric pressure, inches of mercury.
(j) Circulation-air blower speed, rpm (if accessible).

TABLE III. Capacity - airflow test conditions.

| Fuel Type | External Back Pressure Inch Water | Power Input | Preproduction Heater No. | | | | Initial Production Heater No. | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| Type I Heater | | | | | | | | | | |
| DF2 | 0 | 60Hz | X | - | - | - | X | - | - | - |
| DF2 | 0.20 | 60Hz | X | X | X | X | X | X | X | X |
| DF2 | 0.35 | 60Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0 | 60Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0.20 | 60Hz | X | X | X | X | X | X | X | X |
| Gasoline | 0.35 | 60Hz | X | - | - | - | X | - | - | - |
| DF2 | 0 | 50Hz | X | - | - | - | X | - | - | - |
| DF2 | 0.20 | 50Hz | X | - | - | - | X | - | - | - |
| DF2 | 0.35 | 50Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0 | 50Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0.20 | 50Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0.35 | 50Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0.35* | 60Hz | X | - | - | - | X | - | - | - |
| Type II Heater | | | | | | | | | | |
| DF2 | 0 | 400Hz | X | - | - | - | X | - | - | - |
| DF2 | 0.20 | 400Hz | X | X | X | X | X | X | X | X |
| DF2 | 0.35 | 400Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0 | 400Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0.20 | 400Hz | X | X | X | X | X | X | X | X |
| Gasoline | 0.35 | 400Hz | X | - | - | - | X | - | - | - |
| Gasoline | 0.35* | 400Hz | X | - | - | - | X | - | - | - |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

NOTES:

1. Conduct test marked "X" for each heater.
2. Conduct all tests at sea level elevation (0.500 feet).
3. Heater number designations (1, 2, 3, and 4) will be assigned by the Government.
4. Tests marked with "*" shall be conducted utilizing the bottom return air opening with cover plates fastened to the side return air openings.

4.6.2.1.1 Airflow. Airflow measuring apparatus shall be in accordance with AMCA Standard 210. Subject to the approval of the contracting officer, a calibrated, insulated, measuring apparatus utilizing a smooth approach nozzle may be used in lieu of the apparatus specified in AMCA Standard 210. The special apparatus shall contain the components as follows: Straightening vanes; adjustable back-pressure orifice; velocity and static-pressure tubes, thermocouple or therometer points; airflow measuring, smooth approach nozzle; and booster fan. Readings or results obtained by velometers or anemometers are not acceptable.

4.6.2.1.2 Fuel. Fuel weight and consumption shall be obtained with a balance type scale having an accuracy within 0.01 pounds.

4.6.2.1.3 Barometric pressure. Barometric pressure shall be measured with a mercurial barometer and shall be corrected for temperatures of scale and the mercury and for the location of the barometer with regard to altitude and latitude. Readings obtained from aneroid barometers are not acceptable.

4.6.2.1.4 Temperatures. Temperatures shall be measured by appropriately located glass thermometers or thermocouples and calibrated potentiometers. Readings shall be accurate within plus or minus 1/2° F.

4.6.2.1.5 Pressures. Air pressures and pressure differentials shall be measured with liquid manometers having an accuracy of 0.01 inch of water.

4.6.2.1.6 Flue-gas analysis. Flue-gas analysis shall be made with an Orsat as prescribed in ASME Power Test Code, Supplement on Instruments and Apparatus, Part 10-Flue Exhaust Gas Analysis or any electronic analyzer capable of producing readings as small as one thousand parts per million for oxygen and carbon dioxide and as small as one hundred parts per million for carbon monoxide. Electronic analyzers, when used, shall be calibrated and checked with a known value of gas prior to the test. Alternative instruments and procedures for flue-gas analyses shall be approved by the Contracting Officer prior to testing.

4.6.2.1.7 Test results. The test data from the capacity test shall be used to determine the following results:

(a) Airflow, $W_1$ in pound per minute from cfm calculated in accordance with AMCA Standard 210.
(b) Measured output and measured efficiency calculated as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

(1) $Q_1 = (W_1)\ (t_1 - t_2)\ (0.24)\ (60)$

(2) $Q_2 = W_2 \times HHV$

(3) $E_1 = \frac{Q_1}{Q_2} \times 100$

(c) Indicated efficiency, $E_2$, from flue gas losses as determined for all fuels referenced in 3.9 by using flue gas temperature and flue gas analysis data in conjunction with No. 2 fuel oil curves, figure 1, page 14.13, ASHRAE 1977 Fundamentals Handbook, as follows:

$E_2$ = 100 minus percent flue gas losses.

(d) Indicated output from measured input and indicated efficiency.

$$Q_3 = Q_2 \times \frac{E_2}{100}$$

SYMBOLS USED IN CALUCLATIONS

$Q_1$ = measured output, Btu/hr
$Q_2$ = measured input to unit, Btu/hr
$Q_3$ = indicated output, Btu/hr
$E_1$ = measured efficiency, percent
$E_2$ = indicated efficiency, percent
d = air density at temperature measured, lb/cu ft
$W_1$ = Airflow, pounds per minute
$W_2$ = weight of fuel, lb/hr
HHV = higher heating value of fuel, Btu/lb
$CO_2$ = percent carbon monoxide in dry flue gases by volume
CO = percent carbon monoxide in dry flue gases by volume
$O_2$ = percent oxygen in dry flue by volume
$t_1$ = temperature of air leaving unit, °F.
$t_2$ = temperature of air entering unit, °F.
$t_g$ = temperature of exhaust flue gases, °F.

4.6.2.1.8 Basis for failure. One or more of the following shall constitute failure of this test:

(a) Capacity and airflow: The heater has an output heating capacity of less than 60,000 Btu/hr at sea level, or the airflow is not as specified in 3.12.
(b) Flue-gas analysis: The carbon monoxide in the flue gases exceeds 0.02 percent.
(c) Efficiency: The measured efficiency of the heater is less than 65 percent. The indicated efficiency of the heater is less than 70 percent.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

(d) The type I heater utilizing 50 Hertz power malfunctions, does not operate continuously, or does not produce at least 85 percent of the capacity measured during 60 Hertz capacity tests.

4.6.2.2 Altitude. Subject the heater to the following condition for a period of 1 hour: 5,000-foot altitude at an ambient temperature of 60° F plus or minus 10° F. After exposure, and while the heater is at this condition, operate the heater for a period of 1 hour, recording sufficient data to perform the indicated output and efficiency calculations specified in 4.6.2.1.7 at zero inch W.G. external static pressure for both gasoline and diesel fuel. Nonconformance to 3.1 or 3.8 shall constitute failure of this test.

4.6.2.3 Environmental. Each environmental subtest specified herein shall be followed by operation of the heater to determine that the functional ability of the heater and its controls has not been impaired by the environmental exposure. The heater shall be observed while operating in the normal heat mode for 10 minutes, and each control shall be caused to perform each of its functional modes. Failure of any control to be fully functional or observation of abnormal or erratic heater operation shall constitute failure of the preceeding environmental subtest.

4.6.2.3.1 High-temperature. Subject the heater to the following temperatures and operate as indicated:

(a) An ambient temperature of plus 155° F for 4 hours; no solar radiation.
(b) Reduce the ambient temperature, within 1 hour, to 70° F plus or minus 10° F. Operate the heater for not less than 4 hours with both gasoline and diesel fuel (2 hours each).

Nonconformance to 3.6.1 and 3.6.2 shall constitute failure of this test.

4.6.2.3.2 Low-temperature. Subject the heater to the following temperatures and operate the heater as indicated:

(a) An ambient temperature of minus 65° F for 24 hours; no solar radiation. Raise the temperature to minus 50° F. After exposure at minus 50° F for 1 hour, conduct the flame failure test of 4.6.2.4.3. Reconnect the ignition wires while maintaining minus 50° F, start the heater using diesel fuel, class DF-A, having a cloud point not lower than minus 55° F. After heater operation has stabilized, operate the heater continuously for a period of 1 hour and record data to calculate the indicated output and efficiency of the heater. Upon completion of operation using diesel fuel, change to operation on gasoline and operate for 15 minutes. Stop operation and expose the heater to minus 50° F for not less than 2 hours. After exposure and while the heater is at this temperature, start the heater and repeat the operation and recordings specified above using gasoline. Upon



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

completion of the above, for type I heater only, stop operation, expose the heater to minus 50° F for not less than 2 hours and change the power supply frequency to 50 Hz. After exposure and while the heater is at this temperature, start the heater and repeat the operation and recordings specified above.

(b) The low-temperature tests specified in subparagraph (a) above shall be conducted using gasoline and diesel fuel subjected to the same temperature conditions as the heater.

(c) Raise the ambient temperature to 70° F plus or minus 10° F and examine all components for damage or deterioration. Then operate the heater using gasoline for a period of not less than 30 minutes and observe operation.

The following shall constitute failure of this test:

(1) Nonconformance to 3.6.1 and 3.6.2 and 3.13.2.2.
(2) Indicated capacity and efficiency not within 90 percent of that obtained in the tests specified in 4.6.2.1.

Note: The storage and operational portions of the low temperature test may be conducted separately. Perform step (c) above, following the 24 hour period at minus 65° F to complete the storage portion. Begin the operational portion by exposing the heater to minus 50° F for 4 hours prior to performing the flame failure test and commencing operation with DA-4 diesel fuel.

4.6.2.3.3 Humidity. Conduct the humidity test as specified in MIL-STD-810, method 507.1, procedure I. The heater shall not be operated during exposure. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.2.3.4 Salt-fog. Conduct the salt-fog test as specified in MIL-STD-810, method 509, procedure I. The heater shall not be operated during this exposure. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.2.3.5 Vibration. Conduct the vibration test as specified in MIL-STD-810, method 514.2 category f, procedure VIII, table 514.2-VI, curve V. The test time shall be 60 minutes total per plane. The heater shall not be operated during this test. The heater shall be mounted in the down discharge air position, and secured to the vibration machine by its two mounting brackets. Auxiliary braces, straps, or other tiedown devices shall not be employed to secure the heater to the test apparatus. No blocking or tiedown of internal components, other than those intended to remain as part of the heater during its normal operation, shall be employed. Removable panels and access doors shall be in place during all testing. Nonconformance to 3.6.4.2 shall constitute failure of this test.

4.6.2.3.6 Shock. The heater shall be subjected to three shocks perpendicular to each of the six faces (four sides, top and bottom - a total of 18 shocks). Each shock shall be of not less than 15 g's and 11-milli-second duration, and shall conform to figure 516.2 of MIL-STD-810, method 516.1. The



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

heater shall not be operated during this test. Conduct the shock test with the heater in the air discharge down position, mounted by means of its two mounting brackets. Auxiliary braces, straps or other tiedown devices shall not be employed to secure the heater to the test apparatus. No blocking or tiedown of internal components other than those intended to remain as part of the heater during its normal operation shall be employed. Nonconformance to 3.6.4.1 shall constitute failure of this test.

4.6.2.4 Controls.

4.6.2.4.1 Heat output control. Subject the room thermostat to the following test. Set the indicator at the minimum temperature setting. Connect the thermostat to the heater and record the ambient temperature in the vicinity of the thermostat. Place the on-off switch to the "ON" position. Increase the indicator setting until the heater becomes operable. Record the indicator setting temperature. After 5 minutes of operation, decrease the indicator setting until the heat cycle shuts off. Again, record the indicator setting temperature. Nonconformance to 3.13.1 shall constitute failure of this test.

4.6.2.4.2 High temperature safety. The heater discharge airflow shall be partially blocked during heater operation until the high temperature safety control stops the flow of fuel to the combustion chamber. After the air temperature reaches 260° F the rate of blocking shall be adjusted to limit the rate of average temperature change to 2° F per minute. The temperature of the discharge air shall be recorded when the safety control deactivates the fuel supply. Nonconformance to 3.13.2.1 shall constitute failure of this test.

4.6.2.4.3 Flame-failure safety. Disconnect the ignition wires. Start the heater with the ignition wires still disconnected and begin recording the time it takes to stop the flow of the fuel to the fuel to the combustion chamber. After the control stops the fuel flow, attempt to restart the heater by depressing the manual reset control. Nonconformance to 3.13.2.2 shall constitute failure of this test.

4.6.2.4.4 On-off switch. Place the switch in the "FAN" position. Determine that only the circulation-air blower is operating. Then place the switch in the "ON" position. Determine that the combustion cycle and heater control system is energized. After 5 minutes of operating, place the switch in the "OFF" position. Determine that the combustion cycle and the heater control system is de-energized after purge cycle. Nonconformance to 3.13.3.1 shall constitute failure of this test.

4.6.2.4.5 Fuel control. Disconnect the electrical wire from the heater to the remote fuel pump. Connect the remote fuel pump electrical wire to a separate power supply and operate the heater for 15 minutes. Shut the heater off and allow the fuel pump to operate for 15 minutes. Nonconformance to 3.9 shall constitute failure of this test.

4.6.2.5 Sound level. The heater shall be operating at rated capacity in the discharge down position. Measure the sound level at eight evenly spaced points



Source: https://assist.dla.mil -- Downloaded: 202610-17T18:17Z
Check the source to verify that this is the current version before use.

65 inches above ground level on a circle of 3 foot radius from the geometric center of the heater in accordance with MIL-STD-1474, for steady-state noise. Nonconformance to 3.6.6 shall constitute failure of this test.

4.6.2.6 Operating position variation. Perform the individual performance test specified in 4.6.2.12 for each of the following operating positions:

(a) Air discharge down
(b) Air discharge up
(c) Air discharge left
(d) Air discharge right

Failure of the test specified in 4.6.2.12 shall constitute failure of this test.

4.6.2.6.1 Tilted-operation. While the heater is set-up in each position, required by 4.6.2.6, operate the heater with each edge raised so that the bottom is at an angle of 10 degrees from the horizontal. The operating period for each tilt (four tilts per position) shall be not less than 10 minutes. Nonconformance to 3.6.3 shall constitute failure of this test.

4.6.2.7 Carrying handles. Lift the heater vertically in the attitude of figure 1 and hold in suspension using slings attached to the handles. Determine force exerted on each carrying handle. Increase the force equal to 2-1/2 times the force measured on each handle. Hold each load for 90 seconds. Failure or permanent deformation of the heater or its carrying handles shall constitute failure of this test.

4.6.2.8 Endurance. Operate each of three preproduction heaters for not less than 2,000 hours actual heating time under the conditions specified in 4.6.2.8.2. Each heater shall be operated for not less than 250 hours at each of the power conditions specified in 4.6.2.15 for the applicable type heater. The remainder of the 2,000 hour test for type I heaters shall be run using 120 volt, 60 Hertz, single phase power and for type II heater shall be run using 208 volts, ac, 400 Hertz, three phase power. After completion of the 2000-hour endurance test, the unit shall be subjected to the test specified in 4.6.2.1, at the condition of 0.2 inch back pressure and using gasoline fuel supply. Following the capacity test, the heat exchanger shall be removed from the heater and subjected to the tests specified in 4.6.2.11.

4.6.2.8.1 Records. The following data shall be recorded at the beginning of the test and every 60 hours thereafter, plus or minus 10 hours:

(a) Fuel rate, pounds per hour.
(b) Flue gas temperature, ° F.
(c) Flue gas analysis percent by volume of $CO_2$, $O_2$, and CO.
(d) Power input characteristics, volts, watts.

Records shall be maintained, including full explanations of all servicing including time required, parts replacement, and actual operating time.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

4.6.2.8.2 Endurance test conditions. The test shall be conducted at any ambient temperature between 50° and 70° F. The heater shall be set up on an automatic test stand designed to operate the heater at rated capacity under a cycle operation of 20 minutes on and 10 minutes off. Both diesel fuel and gasoline specified in 3.9 shall be used during this test with the total burning time divided equally. Fuel shall be changed every 500 hours of burning time. Fuel shall be supplied to the heater by a fuel pump conforming to MS51321-1. If at any time the data specified in 4.6.2.8.1 varies by more than 10 percent from that obtained at the beginning of the test the applicable heater components shall be given a detailed inspection to determine the cause of the variations. Controls, except the thermostats, shall be tested every 500 hours as specified in 4.6.2.4.

4.6.2.8.3 Failure criteria. One or more of the following shall constitute failure of this test.

(a) Nonconformance to 3.6.7.
(b) Controls failing test specified in 4.6.2.4.
(c) Heat output not within 90 percent of initial capacity under 4.6.2.1.
(d) Heat exchanger failing test specified in 4.6.2.11.

4.6.2.9 Reliability. Using the MTBF specified in 3.26, the heater shall be tested as specified herein with accept/reject criteria in accordance with MIL-STD-781, Test Plan VII C. The heaters shall be accepted if the test time and failures fall within the "accept" region to the right of the "accept and continue test" region of Test Plan VII C, and rejected if test time and failures fall within the "reject" region. If the assessment falls within the "accept and continue test" region, additional testing as specified in 4.6.2.9.1 shall be performed until an accept/reject decision is reached.

4.6.2.9.1 Procedure. Subject the heaters to the test specified in 4.6.2.8, using only gasoline conforming to VV-G-1690 and only 60 Hertz power.

4.6.2.9.2 Failure criteria. A failure is defined as any malfunction which the operator cannot remedy by adjustment, repair or replacement action using the controls, parts, and tools supplied with the units within 30 minutes and which causes, or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance below designated levels.
(b) Heater output less than 90 percent of required capacity.
(c) Controls failing the test specified in 4.6.2.4.
(d) Any impending malfunction which constitutes a safety hazard to operating personnel or would cause damage to the equipment if continued in operation.

4.6.2.9.2.1 Secondary failures. Those malfunctions occuring as dependent or secondary failures shall be excluded from consideration as chargeable failures.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

4.6.2.10 Electromagnetic interference. The heater shall be tested to determine conformance with 3.18. The contractor shall furnish the contracting officer a plan of tests and a report of tests as required by MIL-STD-461. Disapproval of the plan or the report shall contitute failure of this test.

4.6.2.11 Pressure. Remove the heat exchanger from the heater case. The initial pressure tests of each heat exchanger may be conducted prior to its installation into the heater. Seal all external openings of the heat exchanger except one which shall be fitted with an air valve. Subject the heat exchanger to an internal air pressure of 5 psig and submerge the heat exchanger in water. Any evidence of air leakage shall constitute failure of this test.

4.6.2.12 Individual performance. Operate the heater in any ambient temperature between 50° F and 80° F for not less than 30 minutes, using diesel fuel (DF-2), at zero static pressure external to the heater. During this time, make adjustments to obtain proper nozzle pressure and fuel consumption rate and perform control tests as specified in 4.6.2.4. After this period, operate the heater under the same conditions for an additional period of not less than 15 minutes, during which time record not less than the following:

(a) Fuel weight, pounds, for the total 15-minute run. Obtain weight as specified in 4.6.2.1.2. (Continuous measurement by volume for the total period may be used and then converted to pounds).
(b) Flue-gas temperature, °F (read within 1 inch of where flue-gas analysis sample is taken).
(c) Flue-gas analysis, percent $CO_2$, $O_2$, and CO by volume (samples taken from heater exhaust stack at a point 18 inches from the heat exchanger), in accordance with 4.6.2.1.6.

Immediately after completion of this test, each heater being prepared for delivery shall be purged of fuel.

4.6.2.12.1 Basis for failure. One or more of the following shall constitute failure:

(a) The carbon monoxide in the flue gases is more than 0.02 percent.
(b) Indicated output is less than 85 percent of that obtained on the preproduction initial capacity test.
(c) Indicated efficiency less than 70 percent.
(d) Controls failing test specified in 4.6.2.4.

4.6.2.13 Motor locked rotor test. Each motor shall be removed from the heater and a fixture shall be applied to prevent the rotor from rotating. The ambient temperature shall be maintained at 70° F plus or minus 10° F, and rated voltage shall be applied to the motor terminals for 5 hours. After the 5 hour period, the motor shall be de-energized and allowed to cool to room temperature. The rotor locking fixture shall be removed and the motor shall be re-energized and operated for 12 hours. Nonconformance to 3.11.1 or 3.12.1 shall constitute failure of this test.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

4.6.2.14 Combustion safety.

4.6.2.14.1 Combustion air failure. Operate the heater for a period of not less than 30 minutes and, while operating, stop the fan suppling air for combustion. If the flame is extinguished following interruption of the air supply, the air supply shall be restored and the heater shall be restarted. If combustion continues following interruption of the air supply, allow the burner to continue firing in that manner for not less than 4 hours. At the end of 4 hours the combustion air supply shall be restored. Following either above conditions of operation, subject the heater to the individual performance specified in 4.6.2.12. Failure of the test specified in 4.6.2.12, or expulsion of flames from the heater on re-ignition shall consitute failure of this test.

4.6.2.14.2 Interruption of fuel. While the burner is operating, interrupt the fuel to the burner. Allow the heater to continue operation through the purge cycle and until the flame-failure control causes automatic shutdown. Resume heater operation by restoring fuel supply to the burner and resetting the flame-failure control. The following shall constitute failure of this test:

(a) More than 60 seconds expire between the end of the purge cycle and automatic heater shutdown.
(b) The heater fails to start and operate following restoration of fuel and resetting of the flame-failure control.

4.6.2.15 Voltage and frequency variation. Perform the individual performance test specified in 4.6.2.12 for each of the following electrical power input characteristics:

Type I heater

(a) 103.5 volts, 60 Hz.
(b) 126.5 volts, 60 Hz.
(c) 110 volts, 50 Hz.
(d) 120 volts, 50 Hz.

Type II heater

(a) 188 volts, 400 Hz.
(b) 228 volts, 400 Hz.
(c) 198 volts, 380 Hz.
(d) 198 volts, 420 Hz.
(e) 218 volts, 380 Hz.
(f) 218 volts, 420 Hz.

The indicated capacity requirements may be varied directly as the electrical frequency. Failure of the test specified in 4.6.2.12 shall constitute failure of this test.

4.6.2.16 Safety evaluation. Evaluation shall be accomplished throughout testing. Nonconformance to 3.7 shall constitute failure of this test.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

4.6.2.17 Human factors. The heater shall be evaluated to determine that the human factors have been met. Nonconformance to 3.25 shall constitute failure of this test.

4.6.2.18 Maintenance evaluation. Maintenance operations required during initial production tests shall be accomplished as specified in 3.4. Any assemblage or support equipment such as technical manuals, repair parts, special tools, etc., provided under the contract for operation or conducting maintenance will be evaluated. Failure of the heater to be maintained as specified in 3.4 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select heaters at any time during the contract production period and subject these heaters to the examination specified in 4.6.1 and to the tests marked "X" in column 4 of table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production heaters. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison heaters and shall be cause for refusal by the Government to continue acceptance of production heaters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on heaters previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all heaters accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The preproduction pack of the heaters for level A, or B shall be subjected to the drop test in accordance with FED-STD-101, test method 5007, procedure E. At the conclusion of the preproduction pack test, the pack and all contained unit packs shall be opened and examined along with their contents. Rupture of the pack, unit pack, or any visible damage to the content shall constitute failure of the preproduction pack and shall be cause for rejection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the characteristics listed in table IV. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE IV Examination Schedule - Packaging.

| Number | Characteristic | Requirement Paragraph |
|---|---|---|
| Major | | |
| 130. | Preproduction pack not as specified. | 5.1 |
| 131. | Preservation not level A or Commercial as specified (see 6.2). | 5.2 |
| 132. | Each heater not preserved for level A as specified. | 5.2.1.1 |
| 133. | Thermostat not preserved for level A as specified. | 5.2.1.1.1 |
| 134. | Electrical connectors not preserved for level A as specified. | 5.2.1.1.2 |
| 135. | Cover plates and deflector hood not preserved for level A as specified. | 5.2.1.1.3 |
| 136. | Remote fuel pump not preserved for level A as specified. | 5.2.1.1.4 |
| 137. | Consolidation for level A not as specified. | 5.2.1.2 |
| 138. | Each heater not preserved for Commerical in accordance with MIL-STD-1188. | 5.2.2 |
| 139. | Packing not level A, level B, or Commercial as specified (see 6.2). | 5.3 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

TABLE IV Examination Schedule - Packaging. (Cont'd)

| Number | Characteristic | Requirement Paragraph |
|---|---|---|
| 140. | Each complete heater not packed for level A as specified. | 5.3.1 |
| 141. | Each complete heater not packed for level B as specified. | 5.3.2 |
| 142. | Each heater not packed for Commercial in accordance with MIL-STD-1188. | 5.3.3 |
| Minor | | |
| 201. | Repair parts not preserved for level A as specified. | 5.2.1.1.5 |
| 202. | Maintenance tools not preserved for level A as specified. | 5.2.1.1.6 |
| 203. | Technical publications not preserved for level A as specified. | 5.2.1.1.7 |
| 204. | Marking for military levels of protection not in accordance with MIL-STD-129. | 5.4.1 |
| 205. | Marking for commercial packaging not in accordance with MIL-STD-1188. | 5.4.2 |

5. PACKAGING

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production preservation and packing that the applied preservation, packing, and marking comply with the preparation for delivery requirements of this specification (see 6.5). Examination and tests shall be those specified herein.

5.2 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Heater. Each heater, completely drained of all fuel, and all the other components specified herein that are procured with the heater, shall be



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

placed in a close-fitting box conforming to PPP-B-636, V3c. The components shall be immobilized within the box in a manner to prevent movement or damage by the use of fiberboard pads or cut shapes, fabricated from fiberboard conforming to PPP-F-320, class weather resistant, variety and grade optional. The box shall be closed and sealed as specified for Method V in the appendix to the box specification.

5.2.1.1.1 Thermostat. The theromstat shall be preserved in accordance with MIL-P-116, method IC-2, using the contractor's carton or a close-fitting box conforming to PPP-B-566.

5.2.1.1.2 Electrical connectors. The electrical connectors shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.1.3 Cover plates and deflector hood. The cover plates and deflector hood shall be wrapped with the material used for fabricating the method IC-1 or IC-3. The material shall be secured in place with tape conforming to PPP-T-45, type, class, and grade optional.

5.2.1.1.4 Remote fuel pump. The remote fuel pump shall be preserved in accordance with MIL-P-116, method IC-2, using the contractor's carton or a close-fitting box conforming to PPP-B-566 or PPP-B-636.

5.2.1.1.5 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.1.6 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.2.1.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.2 Consolidation. When, at the option of the contractor, it is more convenient to consolidate the preserved components in order to accomplish the immobility specified in 5.2.1.1, the components shall be consolidated in a close-fitting box conforming to PPP-B-636, W5c or W6c, and secured to prevent movement or damage.

5.2.2 Commercial. Each heater completely drained of all fuel, shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.3.1 Level A. Each complete heater, preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification. The strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable, and unless otherwise specified shall be finish B (see 6.2). When specified, finish A strapping shall be used.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

5.3.2 Level B. Each complete heater, preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-601, Domestic type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.3.3 Commercial. Commercial packing of the heater shall be in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The space heaters are intended for use in providing heated air to maintain adequate working temperatures in military vans and temporary shelters.

6.2 Ordering data. Procurement documents should specifiy the following:

(a) Title, number, and date of this specification.
(b) Type of heater required (see 1.2).
(c) Mounting position required if other than as specified (see 3.1).
(d) Time frame required for submission of preproduction models, see 3.2).
(e) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(f) When initial production inspection is required, and number of heaters to be furnished when applicable (see 3.3).
(g) When remote fuel pumps are required (see 3.9.1).
(h) When a deflector hood is required (see 3.12).
(i) Color required (see 3.24).
(j) When the welding process shall not be at the option of the contractor (see 3.27.6).
(k) When a preproduction pack is not required (see 5.1).
(l) Time frame required for submission of preproduction pack, when applicable (see 5.1).
(m) Preservation packing required (see 5.2 and 5.3).
(n) When other than finish B strapping is required (see 5.3.1).

6.3 Preproduction models. Any changes or deviations of production heaters from the approved preproduction models during production will be subject to the



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

approval of the contracting officer. Approval of the preproduction models will not relieve the contractor of his obligation to furnish heaters conforming to this specification.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the heater covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the heaters in accordance with this specification.

6.6 Electromagnetic interference.

(a) This heater has been previously procured and qualified to MIL-STD-461A, Notice 1, and MIL-STD-462, Notice 1, as specified in Section 3 and 4 of this specification. Subsequent notices or revisions should not be made applicable to a procurement contract.

(b) The contracting officer should include the following in procurement documents:

"(1) Notices or revisions other than those listed below are not applicable:

MIL-STD-461A, Notice 1
MIL-STD-462, Notice 1

"(2) Bidders are cautioned against failing to properly take into consideration the radio frequency interference suppression/electromagnetic interference characteristics requirements for the end items in preparing their bids."

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractors maintenance and operation manual to be furnished with each heater.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of heaters.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Preparing activity:
Army - ME

Project 4520-0265

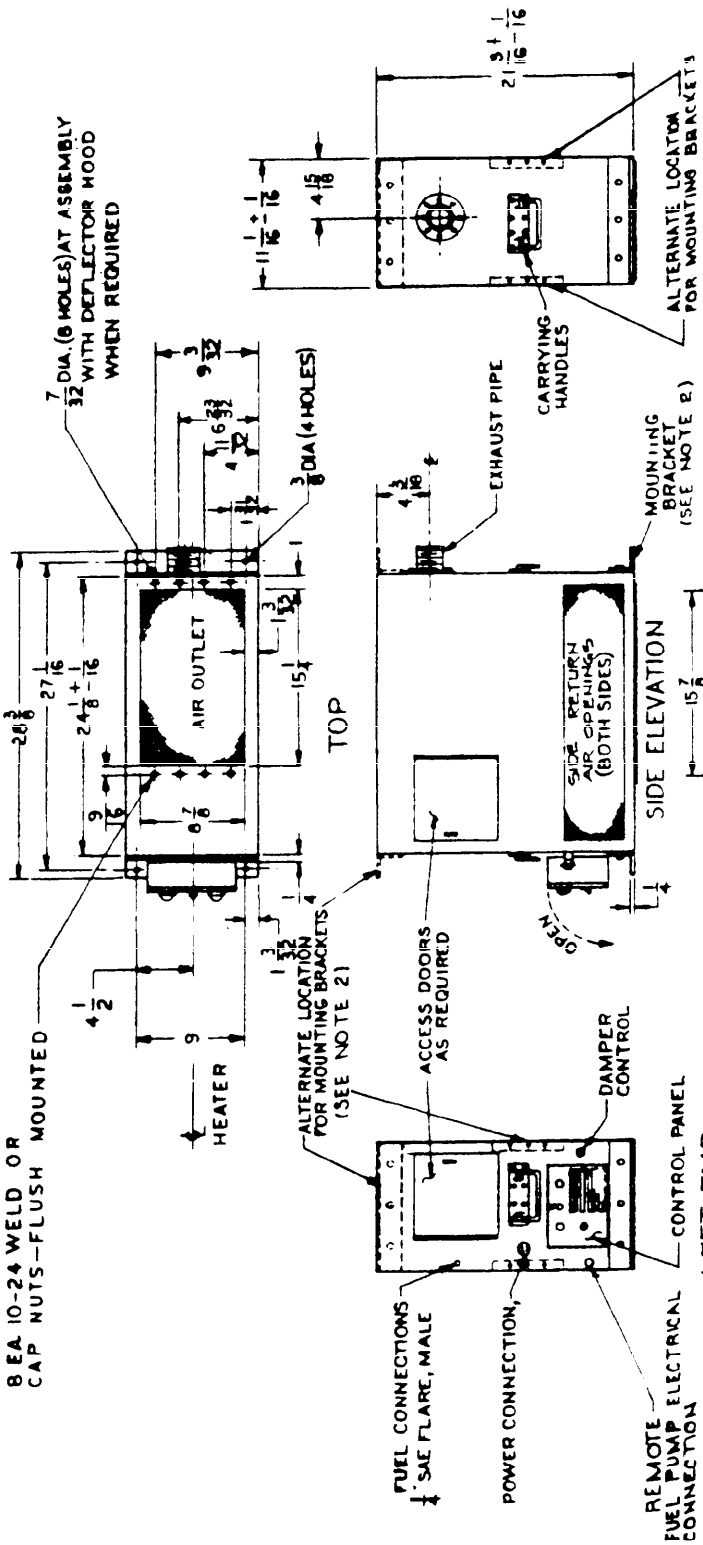
Review activities:
Army - MI, CR
CLA - CS

User activity:
Navy - MC

☆ U.S. GOVERNMENT PRINTING OFFICE: 1980—603-121/3307



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.




NOTE

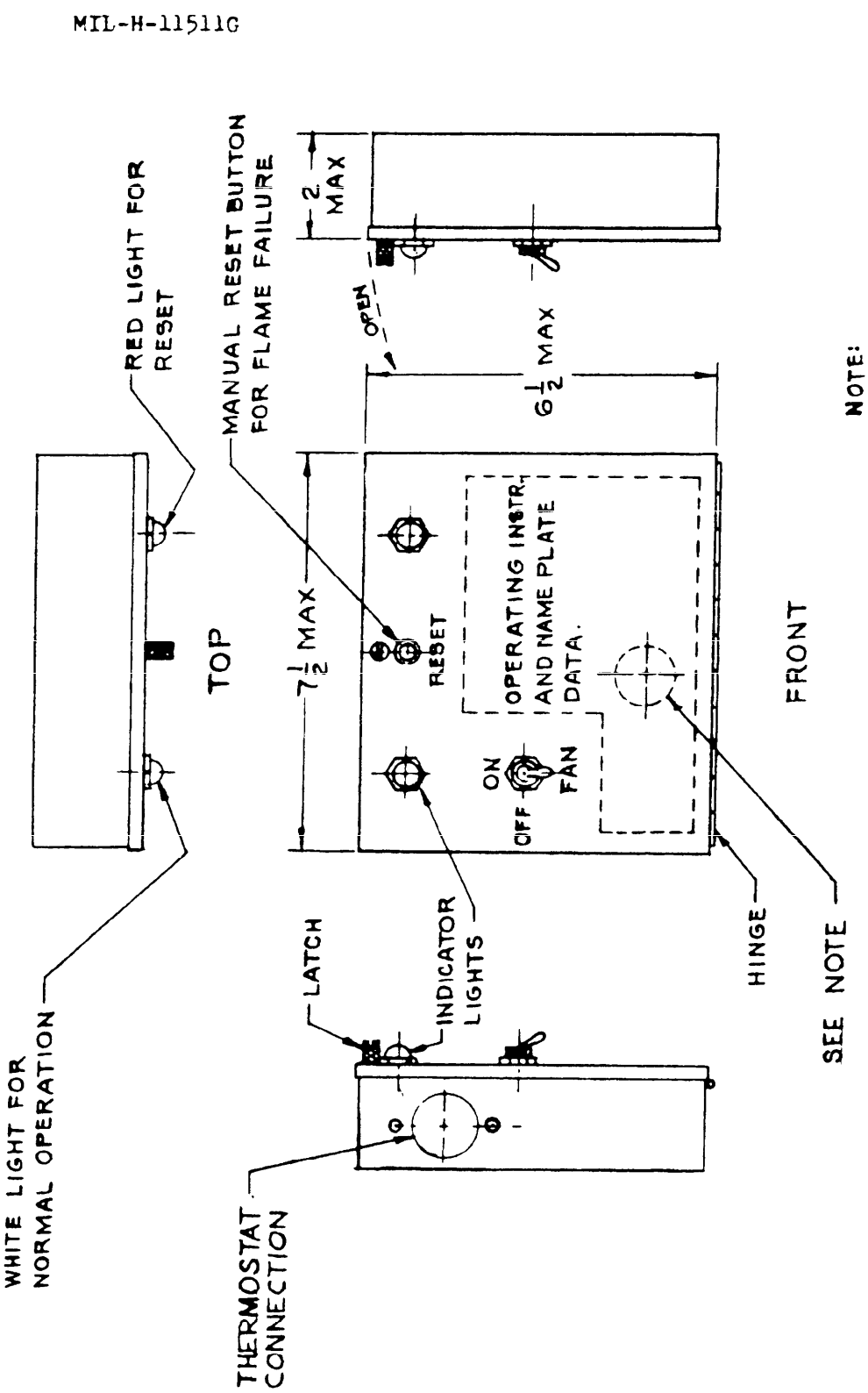
1. LOCATION OF POWER & FUEL CONNECTIONS ON THIS END MAY BE VARIED IF NECESSARY
2. MOUNTING BRACKETS & HEATER SHOWN IN DISCHARGE-UP POSITION. ALTERNATE LOCATION FOR MOUNTING BRACKETS ARE FOR DISCHARGE-DOWN, HORIZONTAL-DISCHARGE LEFT OR HORIZONTAL DISCHARGE-RIGHT POSITIONS.
3. DIMENSIONS IN INCHES.
4. TOLERANCES ± 1/32 EXCEPT AS NOTED.

FIGURE 1 Heater, space, with blower, 60,000 btu/hr (discharge up position) (see note 2).

X-1028C



Source: https://assist.dla.mil -- Downloaded: 202610-17T18:17Z
Check the source to verify that this is the current version before use.




NOTE:
PROVIDE SUITABLE PLUG & RECEPTACLE FOR CONNECTING TO HEATER CIRCUIT.

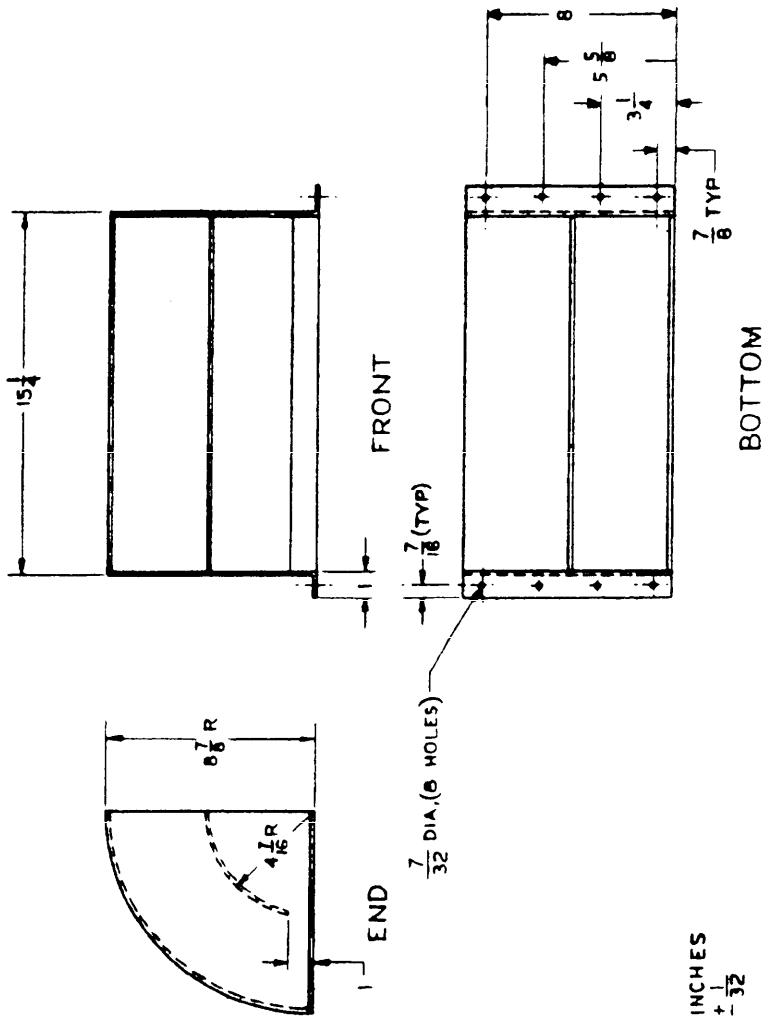
X-2210

FIGURE 2
HEATER CONTROL BOX



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

CX 1029 4




NOTES:
1. DIMENSIONS IN INCHES
2. HOLE LOCATIONS ± 1/32

FIGURE 3. DEFLECTOR HOOD



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-H-11511G Heaters, Space: Multifuel, with Blower 60,000 BTU/Hr

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
USA Mobility Equipment R&D Command
ATTN: DRDME-DE/DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:17Z
Check the source to verify that this is the current version before use.